MultiChartQA-R: A Benchmark for Multi-Chart Question Answering in Real-World Reasoning Scenarios

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Abstract

Existing multi-chart question answering datasets have provided a valuable foundation, but further resources are needed to better study the capabilities of multimodal large language models (MLLMs) in realistic reasoning scenarios. To this end, we present **MultiChartQA-R**, a dataset for multi-chart question answering that extends existing resources with four progressively more complex reasoning tasks, covering cross-chart trend comparison, complementary data integration, anomaly and causal analysis, and strategy recommendation. We further develop a data construction pipeline to support scalable dataset creation and extension to additional languages. In addition, we introduce tailored evaluation protocols for both multiple-choice answering and generative reasoning. Experiments on a diverse set of representative MLLMs show substantial gaps from human performance, particularly in cross-chart visual perception, data integration, and preference-aligned decision-making. We further use an extended dataset to study retrieval and scaling behavior as the number of charts increases. Additionally, our experiments reveal interesting multilingual characteristics of multi-chart question answering. Code and data are available at MultiChartQA-R Hub.

CCS Concepts

• Computing methodologies → Computer vision; **• Information systems → Multimedia information systems**.

Keywords

multi-chart question answering, multimodal reasoning

1 Introduction

Multimodal large language models (MLLMs) have recently shown strong performance on a wide range of vision-language tasks, including visual question answering (VQA) [12, 18, 32], chart-to-code generation [41], image captioning [1, 16, 30], and chart question answering [20, 24, 26, 35, 43]. Among these tasks, chart question answering is particularly important for real-world analytical workflows, where charts are widely used to summarize trends, patterns, and relationships in data. In practice, however, many decision-making scenarios require reasoning over multiple charts rather than a single chart in isolation. For example, analysts often compare multiple stock-related charts to assess market dynamics, researchers examine several experimental plots to identify patterns, and business managers jointly inspect charts on performance and cost before making strategic decisions. Despite these practical needs, the capabilities of MLLMs in realistic multi-chart reasoning scenarios remain underexplored.

Existing chart analysis benchmarks primarily focus on single-chart settings [14, 15, 24, 26, 35, 39], with an emphasis on data extraction and reasoning within a single visualization. These benchmarks do not sufficiently capture the cross-chart evidence integration, explanation, and decision-making required in real-world multi-chart analysis. While MultiChartQA [45] provides an important first step toward multi-chart QA, its task coverage remains centered on information extraction and comparison, with limited support for deeper reasoning such as anomaly diagnosis and strategy recommendation. In addition, existing multi-chart resources are largely English-centric, which limits their utility for multilingual analytical settings.

To address these limitations, we introduce **MultiChartQA-R** (fig. 1), a benchmark designed to evaluate multi-chart question answering from foundational reasoning skills to decision-oriented applications. **MultiChartQA-R** defines four progressively more complex tasks (section 2.1): **cross-chart trend inference**, **complementary data integration**, **anomaly and pattern analysis**, and **strategy recommendation**. Together, these tasks form a coherent progression from trend-level reasoning to decision-oriented synthesis.

Beyond the benchmark design itself, we present the data construction pipeline of MultiChartQA-R (section 2.2), including chart-code pair construction, QA synthesis, and multilingual expansion. The benchmark encompasses 14 chart categories across 36 domains, with English, Chinese, and Spanish versions currently available. Each language version contains 2,160 QA pairs. We further summarize the dataset statistics, quality control procedures, and comparisons with prior benchmarks in section 2.3. To evaluate model performance on the more complex decision-oriented tasks, we introduce a strict risk-aware metric together with a generative evaluation protocol for Task 3 and Task 4 (section 3.2). Experiments on a diverse set of representative MLLMs reveal substantial gaps from human performance and highlight persistent challenges in cross-chart perception and data integration (sections 3.3 and 4). They also reveal a distinct multilingual phenomenon: in multi-chart question answering, model performance is more sensitive to changes in prompt language than to changes in chart-text language.

Our main contributions are as follows:

- We introduce **MultiChartQA-R**, a multilingual benchmark for multi-chart question answering with four progressively complex reasoning tasks, covering trend inference, data integration, anomaly/pattern analysis, and strategy recommendation.
- We develop a scalable data construction pipeline for multi-chart QA, including chart-code pair reconstruction, task-specific QA synthesis, and multilingual expansion, enabling broader coverage of realistic analytical scenarios.
- We propose a strict risk-aware evaluation protocol for decision-oriented multi-option tasks and extend it to a generative setting

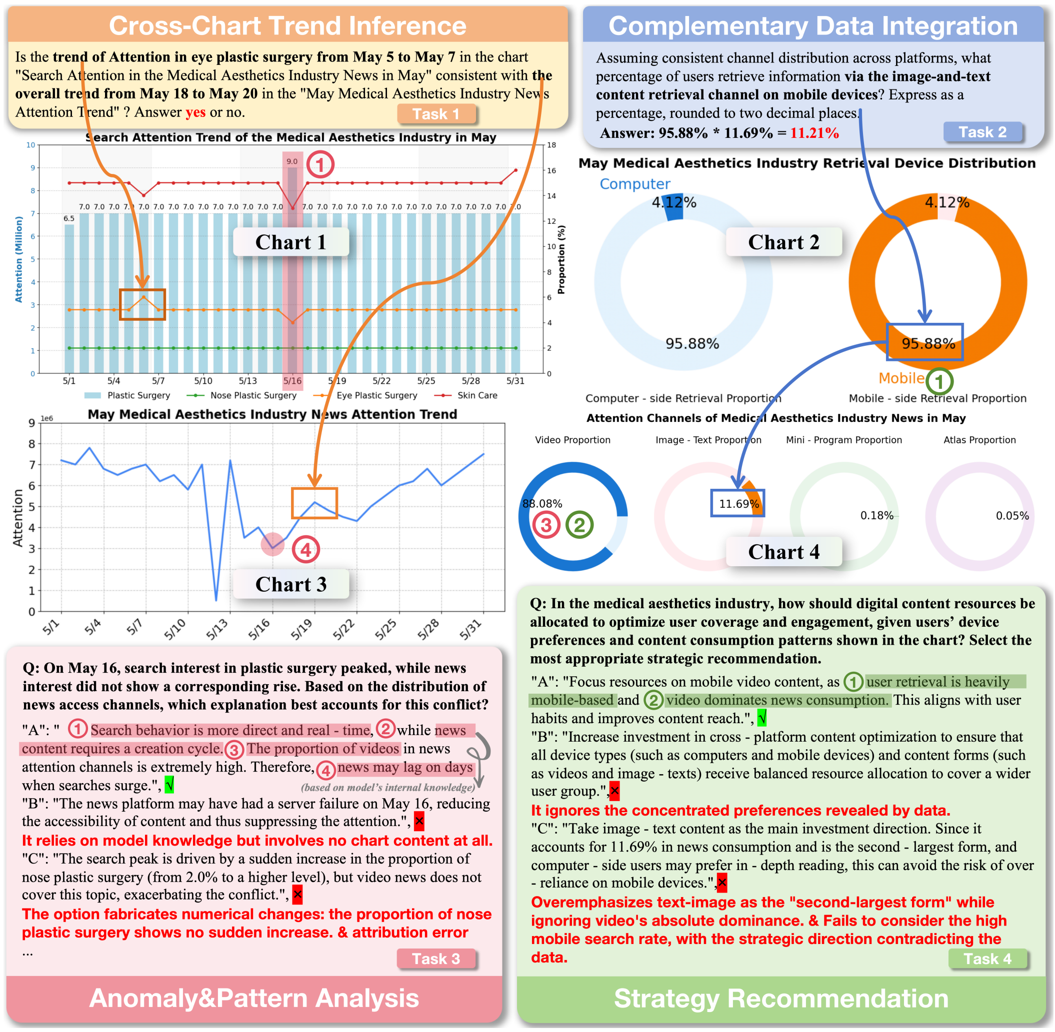

Figure 1: Examples of the four tasks in MultiChartQA-R. Task1&2 use arrows to illustrate the solution process, where the model must identify corresponding trends in the charts for comparison or find complementary data for integration. Task3&4 require identifying related information across multiple charts and reasoning based on the model's internal knowledge to derive the correct inference. Information used by the model is numbered, with erroneous inferences highlighted in red.

for Task 3 and Task 4, enabling more comprehensive assessment of answer correctness and explanation quality.

2 MultiChartQA-R:

In this section, we first introduce the four tasks in MultiChartQA-R (section 2.1), and then describe the data construction pipeline (section 2.2). We next summarize the dataset statistics, quality control procedure, and benchmark comparison (section 2.3).

2.1 Task Definition

As illustrated in Figure 1, MultiChartQA-R includes four tasks that capture progressively more complex forms of multi-chart reasoning, from trend inference to decision-oriented synthesis.

Cross-Chart Trend Inference. Task 1 evaluates whether the model can determine the relational trend between indicators appearing in different charts. The input consists of multiple charts and a question asking whether two or more trends are aligned, divergent, or otherwise relationally consistent. The expected output is a short judgment (typically Yes/No). This task primarily tests cross-chart perceptual alignment and trend-level comparison rather than deep world-knowledge reasoning.

Complementary Data Integration. Task 2 evaluates whether the model can combine complementary information distributed across multiple charts to derive a numerical or factual answer that is not directly available from any single chart. Solving this task typically requires extracting values from different charts and performing logical or arithmetic composition. This task primarily tests cross-chart data aggregation and visual-numerical reasoning.

Anomaly and Pattern Analysis. Task 3 requires the model to identify a non-trivial phenomenon across multiple charts, such as an anomaly, inconsistency, or latent pattern, and provide a plausible explanation grounded in chart evidence. External knowledge may be used as auxiliary context, but the explanation must remain anchored in the observed cross-chart evidence rather than generic domain priors alone. This task primarily tests evidence-grounded interpretation and causal or pattern-level reasoning.

Strategy Recommendation. Task 4 requires the model to produce an actionable recommendation based on relational patterns identified across multiple charts. Unlike Task 3, whose goal is explanation or interpretation, Task 4 focuses on decision-oriented synthesis: the model must translate chart-supported evidence into a concrete strategic judgment or recommendation. A valid answer must therefore be not only plausible in domain terms, but also specifically justified by the chart evidence.

2.2 Data Construction Pipeline

Figure 2 illustrates the data construction pipeline of MultiChartQA-R, including chart-code pair collection, question-answer pair construction, and multilingual expansion.

Chart-Code Pair Collection We collect chart examples from publicly available industry reports and search-index dashboards, which often contain multiple interrelated charts for real-world analysis scenarios. Since the underlying raw data of these charts is typically unavailable, we adopt an LLM-assisted reconstruction process to recover chart-rendering Python code from the original charts. The reconstructed code is then refined through iterative human feedback to ensure that the generated charts faithfully preserve the original content and visual patterns. This process enables us to build chart-code pairs that reflect realistic data characteristics while maintaining diversity in chart styles and domains.

Question–Answer Pair Construction We adopt different construction strategies for the four tasks according to their annotation requirements. For **Task 1**, which focuses on trend judgment that can often be recognized by human annotators at a glance, we use direct manual annotation. For **Task 2**, we first use GPT to propose candidate questions based on the chart data, and then ask human annotators to filter and rewrite them into questions involving multi-chart data extraction and reasoning. Annotators further provide the corresponding calculation process and derive the final labels with calculator-assisted verification. For **Task 3** and **Task 4**, which require more complex cross-chart reasoning and are substantially more costly to annotate manually, we employ a pipeline of automatic synthesis followed by human correction. Specifically, we use chart-rendering code, which contains the underlying gold-table information, together with task definitions as input to a frontier reasoning model in a few-shot manner to generate questions, correct options, and explanations of why the correct options are valid, with the exact model configuration provided in the appendix. Based on the generated questions, correct options, explanations, and chart-rendering code, the model further produces incorrect options together with explanations of why they are incorrect. To improve domain relevance, we further incorporate web-retrieved knowledge during generation. The generated incorrect options are designed at two difficulty levels. Easy distractors either do not use chart data or contain clearly incorrect conclusions. Hard distractors, in contrast, involve either correct data interpretation with rigorous reasoning but an incorrect conclusion, or misinterpreted chart data combined with seemingly rigorous reasoning. Human annotators finally review and revise the synthesized data to ensure validity and quality.

Multilingual Expansion We extend MultiChartQA-R to multiple languages at both the chart level and the QA level. For chart construction, we translate the textual elements in the chart-rendering code into target languages and then execute the translated code to generate multilingual chart images. During this stage, web retrieval is used to verify domain-specific terms and improve translation accuracy. For QA construction, we translate question-answer pairs conditioned on the corresponding gold-table content, so that the translated text remains consistent with the chart content and underlying data semantics. In this sense, our multilingual expansion is not based on direct machine translation alone, but on a structured intermediate representation that helps preserve terminology and semantic consistency across charts and QA pairs. We currently provide English, Chinese, and Spanish versions, and the same pipeline can be extended to additional languages.

2.3 Dataset Statistics and Comparison

Table 1 summarizes the core statistics of MultiChartQA-R. It encompasses 14 distinct chart categories across 36 diverse domains, providing broad coverage of real-world multi-chart reasoning scenarios (see Appendix appendix A.1). To ensure data quality, we apply rigorous human verification throughout the construction process: Tasks 1–2 are fully cross-reviewed, while Tasks 3–4 are generated through model-assisted synthesis and then comprehensively reviewed and refined by human annotators before inclusion in the benchmark. We further report the quality inspection protocols and human performance benchmark in Appendix appendix B. As shown in Table 2, MultiChartQA-R extends prior benchmarks in task coverage, multilingual support, data accessibility, and evaluation design.

3 Experiments

3.1 Experimental Setup

We evaluate a diverse set of representative proprietary and open-weight MLLMs on MultiChartQA-R, with model references provided in Table 3. All evaluations use Chain-of-Thought prompting [36], and the prompts are provided in Appendix G.

3.2 Evaluation Metric

Cross-Chart Trend Inference employs strict string matching, as answers are mostly Yes/No.

Complementary Data Integration uses relaxed accuracy [24], allowing a 5% relative error in numerical answers.

Anomaly and Pattern Analysis and **Strategy Recommendation** involve multi-option answers with distractors of unequal severity. In particular, selecting an option that clearly contradicts chart evidence should be penalized more heavily than selecting a semantically plausible but ultimately incorrect alternative. We therefore use the **Strict Risk-Aware MF_β Score**, which applies fixed severity weights to the selected option set.

Let A be the set of correct answers (Ground Truth) and B be the set of answers selected by the model. The incorrect options are

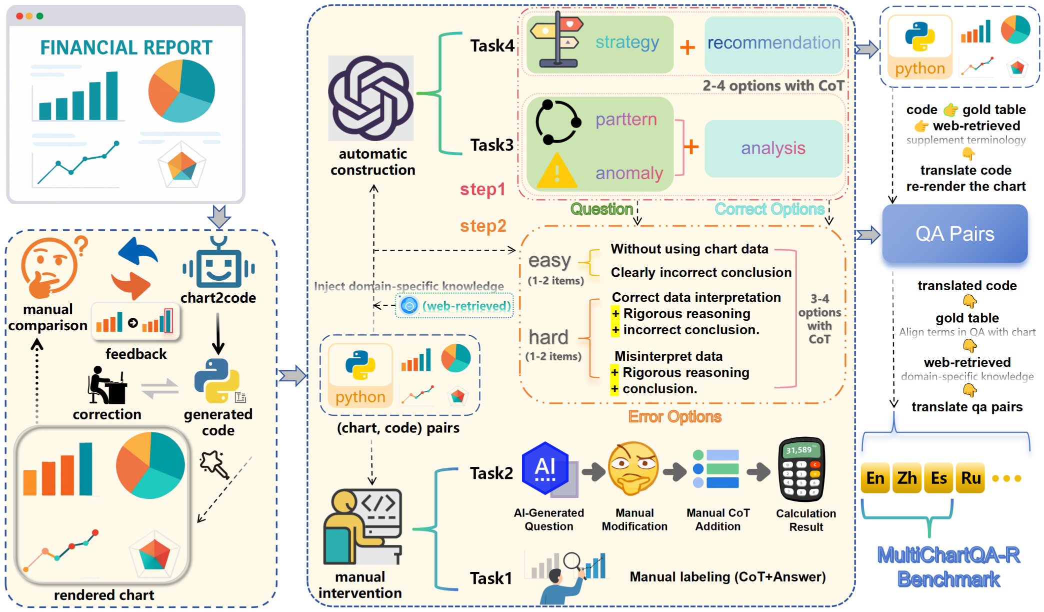

Figure 2: The Construction Pipeline of MultiChartQA-R

Statistic	Number	Category	Number	Ans Type
Total Questions	2160	- Cross-chart Trend Inference	540(25%)	Yes/No
Unique charts	695	- Complementary Data Integration	540(25%)	Generation
Multi-chart sets	180	- Anomaly and Pattern Analysis	540(25%)	Multi-option + Generation
Average charts	3.9	- Strategy Recommendation	540(25%)	Multi-option + Generation

Table 1: Statistics of MultiChartQA-R (shown for one language version). The benchmark covers English (en), Chinese (zh), and Spanish (es); Tasks 3 and 4 support multiple-choice and generative modes.

Benchmarks	Reflect Real Scenarios	Topic Diversity	MultiChart	Multilingual	CoT	Chart-Code Pairs	Gold Table	Evaluation Metric	# of Chart Types
PlotQA [26]	✗	✗	✗	✗	✗	✗	✓	Accuracy	3
ChartQA [24]	✓	-	✗	✗	✗	✗	✓	Accuracy	3
ChartXiv [35]	✓	✓	✗	✗	✗	✗	✗	GPT-4 Score	18
ChartQAPro [23]	✓	✓	✓	✗	✗	✗	✗	Accuracy + ANLS score	9+
MultiChartQA [45]	✓	-	✓	✗	✗	✗	✗	Accuracy	-
MultiChartQA-R(Ours)	✓	✓	✓	✓	✓	✓	✓	Accuracy + MF_β + Gen. Score	14

Table 2: Comparison of MultiChartQA-R with existing chart-based QA benchmarks.

categorized into two subsets: Easy/Obvious Errors (E) and Hard/-Confusing Errors (H). We define the key quantities:

$$TP = |A \cap B|, \quad e = |B \cap E|, \quad h = |B \cap H| \tag{1}$$

The **Risk-Aware Net True Positive** is computed as:

$$TP^{\star}_{net} = \max(0,\ TP - w_e \cdot e - w_h \cdot h) \tag{2}$$

where the severity weights are set to $w_e = 1.0$ and $w_h = 0.5$ in our main benchmark, but can be adjusted to reflect different real-world risk preferences. Here, "easy" and "hard" describe distractor discriminability rather than evaluation importance. We assign a larger penalty to easy errors because they correspond to clearer grounding failures or explicitly incorrect conclusions, whereas hard errors are designed to be more plausible near-miss alternatives.

Based on $TP^{\star}_{net}$, we define **Strict Precision** and standard Recall:

$$P^{\star}_{strict} = \begin{cases} 0, & |B| = 0 \\ \frac{TP^{\star}_{net}}{|B|}, & |B| > 0 \end{cases} \qquad R = \begin{cases} 0, & |A| = 0 \\ \frac{|A \cap B|}{|A|}, & |A| > 0 \end{cases} \tag{3}$$

The final $MF^{strict\star}_{\beta}$ score is:

$$MF^{strict\star}_{\beta} = \begin{cases} 0, & \beta^2 P^{\star}_{strict} + R = 0 \\ (1+\beta^2) \cdot \frac{P^{\star}_{strict} \cdot R}{\beta^2 P^{\star}_{strict} + R}, & \text{otherwise} \end{cases} \tag{4}$$

In the main benchmark, we fix $\beta = 1$, yielding a balanced setting between strict precision and recall. Detailed analysis of β and metric comparisons is provided in Appendix appendix D.1.

Generative Evaluation for Task 3 & 4. To complement the multiple-choice setting, we additionally evaluate Task 3 and Task 4 in a generative setting, where models produce free-text analyses directly from chart images without seeing the candidate options. We first use a judge LLM to extract the option(s) implied by the generated analysis and score answer correctness using the same risk-aware option-level protocol as in the multiple-choice setting. We then assess explanation quality with an LLM judge along four

Model	Trend Inference			Data Integration			Anomaly/Pattern		Strategy Rec	
	en	zh	es	en	zh	es	Multi-select	Generative	Multi-select	Generative
Human		**97.83**			**94.83**		**90.60**	**85.80**	**91.60**	**87.50**
Proprietary Models										
Claude-Sonnet-4 [3]	70.00	75.78	69.33	60.99	59.64	62.98	72.33	**77.6**	77.49	73.1
Gemini-2.5-Pro [33]	**75.06**	**79.33**	**75.11**	65.08	69.35	65.91	74.66	75.5	75.54	**79.7**
Seed1.5-VL [9]	72.44	71.78	67.33	**67.66**	**72.81**	**68.64**	**74.79**	62.8	**78.40**	74.4
GPT-4o [27]	64.21	62.64	59.87	64.83	63.60	64.77	52.06	58.2	56.16	73.2
Open-weight MLLMs										
InternVL3-78B [44]	73.21	68.46	64.66	**67.50**	**70.72**	63.33	**71.80**	64.2	**73.00**	61.8
Qwen3-VL-32B [4]	73.11	**73.94**	**72.67**	64.22	70.31	**66.59**	67.80	**76.3**	68.35	**77.8**
Qwen3.5-9B [2]	**75.24**	71.53	68.78	62.50	62.67	61.56	58.15	49.0	59.92	56.0
Qwen3-VL-8B [4]	62.36	64.44	60.44	56.22	60.31	51.45	66.75	64.2	70.75	73.7
InternVL2-L3-76B [7]	59.91	48.88	59.19	51.59	53.51	51.14	55.26	62.9	58.54	69.7
Qwen2.5-VL-72B [29]	56.25	56.95	53.13	25.40	14.77	21.51	60.02	**70.7**	63.57	76.3
LLaVA-OV-72B [19]	61.33	57.59	53.33	43.02	16.25	22.22	50.37	54.8	52.85	58.4
InternVL2-26B [7]	54.46	58.65	50.11	31.03	28.70	21.95	46.97	47.0	51.22	47.0
Qwen2.5-VL-7B [29]	54.44	54.91	52.22	21.04	19.64	21.14	58.32	61.7	62.21	65.6
MiniCPM-V-2_6 [42]	49.88	55.36	44.22	23.29	26.15	16.91	40.48	54.1	41.23	59.6
DeepSeek-VL-7B [22]	49.32	47.11	40.44	7.95	5.01	6.74	24.61	50.5	29.32	53.0
LLaVA-OV-7B [19]	48.55	34.00	46.00	21.84	8.20	12.38	31.00	49.9	38.77	58.5

Table 3: MultiChartQA-R leaderboard. For Task 3–4, Multi-select reports Strict Risk-Aware MF_β ($\beta = 1$, $w_e = 1.0$, $w_h = 0.5$), and Generative reports the free-form generation score. Best scores are in bold.

dimensions: *Evidence Relevance*, *Reasoning Completeness*, *Hallucination Risk*, and *Consistency*. We further report reference-based similarity metrics as supplementary indicators. Detailed prompts, rubrics, and scoring procedures are provided in Appendix appendix D.7.

3.3 Main Results

Table 3 reveals significant findings regarding current MLLM capabilities in multi-chart scenarios:

Significant Human-AI Gap. Human evaluators consistently achieved much stronger results than current models across the benchmark. In contrast, even strong proprietary models such as Claude-Sonnet-4 and Gemini-2.5-Pro remained well below human performance. This underscores that cross-chart analysis, which requires joint perception and long-chain reasoning, remains a distinct challenge compared to single-image understanding.

Task-Specific Challenges under Multi-Select and Generative Evaluation. Performance varies drastically across task types, highlighting different model deficiencies:

Trend Inference (Task 1) remains challenging for many models. Since the task is binary, scores near 50% indicate near-chance performance, suggesting failures in aligning temporal patterns across multiple charts. Only a subset of models, such as Gemini-2.5-Pro and Qwen3-VL-32B, demonstrate clear cross-chart perceptual ability beyond random guessing.

Data Integration (Task 2) further exposes weaknesses in visual-numerical alignment, as many models struggle to accurately extract and combine values from multiple charts. For the more complex reasoning tasks, Anomaly and Pattern Analysis (Task 3) and Strategy Recommendation (Task 4), the multi-select and generative settings reveal complementary aspects of model behavior. In the multi-select setting, models must distinguish the correct answer from plausible but ultimately incorrect distractors, so low scores indicate difficulty in separating valid cross-chart reasoning from convincing hard negatives. In the generative setting, models must produce free-form analyses without access to candidate options, which more directly tests explanation generation and open-ended reasoning. Taken together, these two settings show that current MLLMs still face substantial challenges in both structured option discrimination and free-form multi-chart reasoning.

Open-Weight Competitiveness and Multilingual Fragility. Several open-weight models, especially Qwen3-VL-32B and InternVL3-78B, achieve performance competitive with widely used proprietary baselines such as GPT-4o, particularly in visual perception, data integration, and parts of the reasoning tasks. This highlights the growing competitiveness of open-weight MLLMs in multi-chart understanding. However, multilingual analysis still reveals clear fragility. While English and Chinese performance is often relatively stable, some models show notable degradation in Spanish. Detailed multilingual comparisons, including the reasoning-oriented settings, are provided in the appendix. These results suggest that robust multilingual multi-chart question answering remains an open challenge for current MLLMs.

4 Discussion

4.1 Impact of Irrelevant Charts

Robustness via Intrinsic Relevance Assessment. We evaluated robustness by introducing topic-related but answer-irrelevant charts alongside the essential ones ('all' vs. 'involved'). Table 6 shows that proprietary models (e.g., Claude-Sonnet-4) remain invariant to these distractors. This stability implies superior intrinsic relevance assessment, enabling them to precisely isolate essential visual data even when surrounded by thematically similar noise—a capability currently exclusive to closed-source systems.

The "Noise Sensitivity" Bottleneck. Conversely, open-weight models display marked "noise sensitivity" to these hard negatives. Despite matching baselines on curated inputs, they suffer sharp degradation in the expanded context (e.g., InternVL3-78B's ~12% drop in Trend Inference). This reveals a deficit in *discriminative perception*: while capable of reasoning with provided data, current open architectures struggle to reject information that is thematically consistent but logically useless. For a detailed ablation study concerning *Visual Dependency and Language Priors*, please refer to appendix B.

4.2 Cross-Lingual Asymmetry

Comparing Table 11 and Table 12 in Appendix appendix E.3 reveals a notable asymmetry in how language shifts affect performance. Models are more sensitive to prompt language than to the language of chart text itself. For instance, in Trend Inference, InternVL2-26B drops by about 8.4 points when the prompt switches from English to Chinese (Table 11), whereas performance changes by only about 0.9 points when the chart text switches to Chinese. This suggests that visual encoders have fairly robust multilingual OCR and semantic alignment capabilities. The main bottleneck therefore lies in cross-lingual instruction following in the reasoning module, rather than in interpreting foreign-language chart text.

4.3 Exploring MLLMs' Retrieval Capabilities

Additionally, we constructed an extended benchmark to analyze MLLMs' retrieval capabilities as the number of charts and the amount of relevant information increase (Appendix appendix F). The results across the four settings (Figure 7 in appendix E.4) show that performance consistently degrades as chart context becomes larger and denser. Interestingly, when extracting one datum per chart, models achieve higher scores by producing a computed result than by directly outputting multiple data points, suggesting that current MLLMs still struggle with parallel retrieval over multiple charts.

4.4 Error Analysis

Our analysis suggests that prompt-format sensitivity is an important factor affecting model performance on the multiple-choice settings of Task 3 and Task 4. In particular, GPT-4o shows a much stronger tendency than Claude-Sonnet-4 and Seed1.5-VL to overfit superficial option patterns rather than reason over chart content. For example, in Task 3, GPT-4o exhibits unusually high misselection rates on options B, C, and E, even though the option order is shuffled across instances. This indicates that the model is overly influenced by recurring answer-format patterns, which leads to higher distractor selection rates and lower overall performance.

In contrast, Claude-Sonnet-4 maintains stronger accuracy with much lower bias toward these option labels, suggesting more balanced option judgment and better suppression of prompt bias [40]. Seed1.5-VL shows slightly lower accuracy than Claude, but still exhibits substantially weaker label bias than GPT-4o. Overall, these results suggest that performance on multi-select tasks is partly constrained by a model's ability to ignore irrelevant structural cues in the prompt and attend to the actual chart-grounded semantics of the options. To minimize such confounds, our main experiments use a neutral prompt template without one-shot answer demonstrations or fixed label patterns; the detailed prompt templates are provided in Appendix appendix G, and the corresponding statistics are reported in Appendix appendix E.5.

5 Related Works

Chart Question Answering Benchmarks Early chart QA benchmarks, such as FigureQA [15], DVQA [14], PlotQA [26], and ChartQA [24], established important foundations for chart understanding by focusing on single-chart data extraction and reasoning. More recent resources, including ChartBench [39], ChartLlama [10], Charxiv [35], and ChartAssistant [25], further expand chart diversity and question complexity, while related tasks such as chart-to-code generation [41] broaden the research landscape. Despite this progress, most existing chart QA benchmarks remain centered on single-chart settings.

Multilingual Chart Question Answering Benchmarks The growth of multilingual VQA benchmarks, such as xGQA [28], MaXM [5], and CVQA [31], has helped reduce the English-centric bias in visual question answering. For charts, POLYCHARTQA [38], OneChart [6], and KITAB-Bench [11] represent valuable early efforts toward multilingual chart understanding. However, multilingual chart QA remains relatively underexplored, and existing benchmarks mostly focus on structural extraction or shallow reasoning rather than more demanding analytical tasks.

Multi-chart Question Answering Benchmarks Multi-image benchmarks such as Mantis-Instruct [13], BLINK [8], and MUIRBENCH [34] study reasoning over multiple images, although they do not specifically target chart data. Existing multi-chart chart benchmarks are also emerging: MMC-Benchmark [21] includes a small subset of multi-chart samples, ReMI [17] covers several multi-chart scenarios, and MultiChartQA [45] is the first benchmark dedicated to multi-chart question answering. These efforts provide an important starting point for studying multi-chart reasoning. Building on this line of work, MultiChartQA-R further explores more realistic multi-chart settings with broader task coverage and multilingual evaluation.

6 Conclusion

We presented **MultiChartQA-R**, a benchmark for multi-chart question answering together with a scalable data construction pipeline. Through four tasks, it evaluates abilities ranging from trend inference and data integration to anomaly analysis and strategy recommendation, reflecting increasingly realistic reasoning demands. We further introduced evaluation protocols for both multi-select and generative settings. Our experiments show that current MLLMs still face substantial challenges in realistic multi-chart reasoning, with clear gaps from human performance in cross-chart perception, evidence integration, and decision-oriented reasoning. Additional analyses on multilingual settings, retrieval scalability, and prompt sensitivity further highlight the complexity of this problem. We hope MultiChartQA-R will serve as a useful foundation for future research on more robust and practically effective MLLMs for real-world multi-chart reasoning.

Repository: https://github.com/tomatoyuan/MultiChartQA-R
Appendix: https://tomatoyuan.github.io/MultiChartQA-R/appendix.html

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Appendix

A benchmark

A.1 Statistics

As summarized in Table 1 in the main text, each language version of MultiChartQA-R contains 180 multi-chart sets, 695 chart-code pairs, and 2,160 QA pairs, with 540 instances for each of the four tasks. On average, each set contains 3.9 charts, reflecting the realistic need to aggregate evidence across multiple visualizations rather than reason over isolated charts. The same construction pipeline is used for the English, Chinese, and Spanish versions, so the benchmark remains structurally aligned across languages while supporting multilingual evaluation.

Beyond scale, MultiChartQA-R is designed to provide broad diversity in both visual form and application context. Figure 3 shows that the benchmark covers 14 chart categories, while Figure 4 shows coverage across 36 domains. This diversity is important for reducing over-specialization to a narrow set of chart styles or topics, and for better reflecting the heterogeneous analytical scenarios encountered in real-world multi-chart reasoning.

Taken together, these statistics show that MultiChartQA-R is not only balanced in task composition, but also broad in chart forms and domains, making it suitable for studying both controlled task difficulty and generalization across realistic multi-chart scenarios.

B Data Quality Inspection Details

To ensure the reliability of MultiChartQA-R, we applied task-specific human verification throughout the dataset construction process and complemented it with a post hoc quality audit and a human performance benchmark.

B.1 Quality Control Mechanisms

Tasks 1 & 2: Full Cross-Review. For *Trend Inference* and *Data Integration*, which have objectively verifiable answers, we conducted full human cross-review over the complete dataset. The annotation team consisted of four members, and each item was independently checked by another annotator in a cyclic peer-review process. The review focused on three aspects:

- **Question validity:** whether the question strictly followed the task definition and was unambiguous;
- **Reasoning correctness:** whether the solution process was logically sound and consistent with the chart evidence;
- **Answer accuracy:** whether the final answer was fully supported by the chart data.

Any identified issues were corrected before the data were finalized.

Tasks 3 & 4: Human Refinement of Model-Assisted Synthesis. For *Anomaly and Pattern Analysis* and *Strategy Recommendation*, we adopted a model-assisted synthesis pipeline to generate candidate question-answer items. However, these outputs were not included directly in the benchmark. Instead, all synthesized samples were comprehensively reviewed and refined by human annotators before inclusion. The human revision process focused on four aspects:

- **Question-type alignment:** whether each question matched the intended task type and required the targeted form of multi-chart reasoning;
- **Validity of correct options:** whether the correct option and its explanation were well grounded in the chart evidence and logically coherent;
- **Effectiveness of distractors:** whether incorrect options were plausible yet ultimately invalid, including both easy negatives and hard negatives;
- **Explanation consistency:** whether the explanations for both correct and incorrect options were faithful to the chart evidence and consistent with the intended reasoning path.

This procedure ensured that the final Task 3 and Task 4 data were human-refined benchmark instances rather than raw model outputs.

Post Hoc Quality Audit for Tasks 3 & 4. After the full human refinement stage, we further conducted an expert quality audit on a randomly sampled 30% subset of Tasks 3 and 4. Evaluators scored each sample on a 10-point scale using the criteria above. The audit results indicate strong data quality: Task 3 achieved an average score of **9.1/10** with an inter-rater agreement of **85%**, while Task 4 achieved an average score of **9.3/10** with an inter-rater agreement of **87%**.

B.2 Human Performance Baseline

To establish a human-level reference and verify task difficulty, the expert group solved the benchmark without access to the ground truth. Human performance reached **97.83** and **94.83** on Task 1 and Task 2, **90.60** and **91.60** in the multi-select setting of Task 3 and Task 4, and **85.80** and **87.50** in the corresponding generative setting. Although human performance consistently exceeded that of current models, the more complex tasks still led to a non-negligible error rate, highlighting the intrinsic difficulty and research value of MultiChartQA-R.

B.3 Annotator Recruitment and Compensation

We recruited four annotators with undergraduate degrees or above in STEM-related fields to ensure sufficient expertise in chart interpretation and logical reasoning.

Compensation. For annotation, review, and human evaluation, annotators were paid at a rate of **$15 per hour**, adjusted according to task difficulty. This rate is substantially higher than the local minimum wage (approximately **$5 per hour**), ensuring fair compensation.

Consent. All annotators were informed of the purpose of the study and the intended use of the collected annotations. They provided consent for their contributions to be used for research purposes and released under the project license.

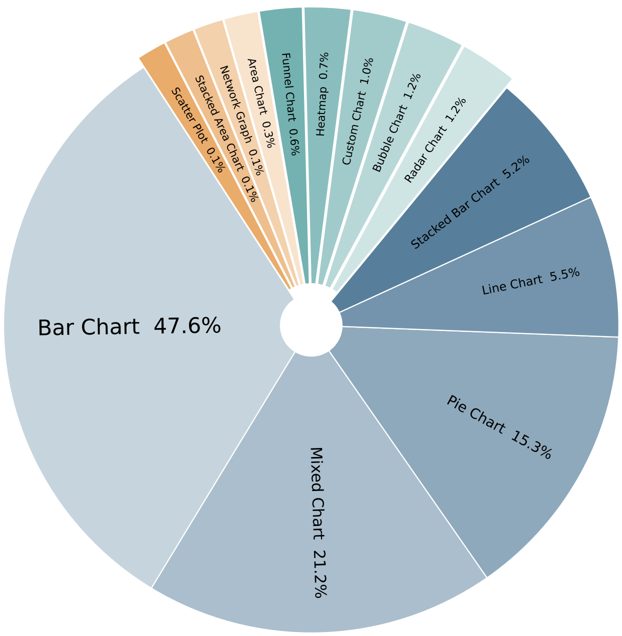

Figure 3: Distribution of Chart Types

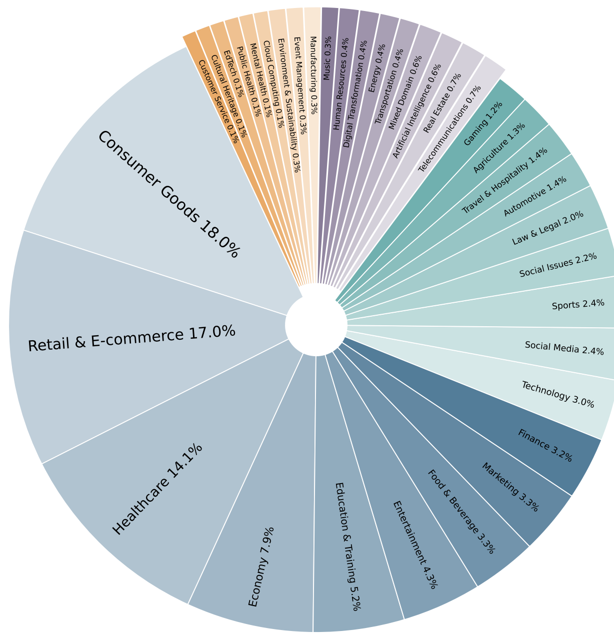

Figure 4: Distribution of Domains

C Experiments

D Metric Exploration and Phenomenon Analysis

To validate the effectiveness of the proposed **Strict Risk-Aware** MF_β **Score**, we conducted a comparative analysis against standard metrics (Original F1) and recent benchmarks (COM2 from ACL 2025) on the English multi-select results of Task 3 and Task 4. The detailed results are presented in Table 4, where the **Strict** column is kept consistent with the main leaderboard.

The "Safety Gap" Phenomenon. A crucial observation from our experiments is the divergence between standard precision-based metrics and our risk-aware metric, which we term the "Safety Gap."

- **High-Performance Models:** For robust models such as Claude-Sonnet-4 and Seed1.5-VL, the gap between ‘Orig F1‘ and ‘Strict‘ is relatively moderate. For example, in Task 4 English, Claude drops from 87.53 to 77.49, a Δ of about 10.0. This indicates that these models not only identify correct answers but also avoid selecting high-risk distractors.
- **Low-Performance Models:** Conversely, weaker models such as DeepSeek-VL-7B exhibit a much larger drop (e.g., in Task 4 English, dropping from 56.95 to 29.32, a Δ of about 27.6).

This phenomenon reveals that standard metrics often inflate the performance of weaker models that engage in "reckless guessing" (hitting some correct options while simultaneously selecting fatal errors). The **Strict** metric successfully penalizes this behavior, offering a more distinct differentiation capability than ‘Orig F1‘ and providing a balanced assessment compared to the overly harsh ‘COM2‘ metric.

Correlation with Human Judgment. As shown in the table, the ranking consistency provided by the ‘Strict‘ metric aligns more closely with the capability tiers observed in manual inspections. While ‘Orig F1‘ might suggest that a 7B model performs comparably to a 72B model in certain scenarios due to high recall, the ‘Strict‘ score exposes the fragility of the smaller model's reasoning, ensuring that high scores truly reflect reliable, deployable decision-making capabilities.

D.1 Evaluation Metric

The original MF_β score (Orig F1) and Com^2 exhibit similar distributions across model evaluations, suggesting that MF_β captures multiple-choice capability reasonably well. However, Com^2 amplifies the impact of distractors, which limits its applicability across scenarios. By contrast, MF_β is more flexible because it can adapt to different settings through the choice of β.

Anomaly and Pattern Analysis and **Strategy Recommendation** involve open-ended multi-chart question-answering, where the answers are not unique. These tasks are designed in a multiple-choice format with Easy and Hard distractors. To evaluate model performance, we propose the **Strict Risk-Aware** MF_β metric with fixed severity weights.

D.2 Design Rationale

The key motivation is that not all false positives are equally problematic in our benchmark. Some distractors are included as *easy errors*: these options either ignore the relevant chart evidence or lead to clearly incorrect conclusions. Selecting such an option indicates a basic failure of chart grounding or decision reliability. By

Model	Task 3 (Anomaly Analysis) - EN			Task 4 (Strategy Rec) - EN		
	Orig F1	**Strict**	COM2	Orig F1	**Strict**	COM2
Proprietary Models						
Claude-Sonnet-4	84.92	72.33	65.04	87.53	77.49	69.67
Gemini-2.5-Pro	81.87	74.66	59.04	83.79	75.54	60.86
Seed1.5-VL	78.87	**74.79**	58.59	82.22	**78.40**	66.15
GPT-4o	71.76	52.06	38.59	76.88	56.16	38.86
Open-Weight Models						
InternVL3-78B	81.62	71.80	57.15	81.78	73.00	57.20
Qwen3-VL-32B	84.64	67.80	63.29	86.63	68.35	64.45
Qwen2.5-VL-72B	72.62	60.02	32.93	76.34	63.57	39.44
InternVL2-L3-76B	68.33	55.26	34.07	70.19	58.54	36.95
Qwen2.5-VL-7B	71.38	58.32	37.33	74.55	62.21	43.19
MiniCPM-V-2_6	60.67	40.48	21.27	60.13	41.23	21.90
DeepSeek-VL-7B	49.90	24.61	11.35	56.95	29.32	17.11

Table 4: Comparison of different evaluation metrics on English multi-select tasks. Orig F1 represents the standard F1 score, which treats all errors equally. COM2 is a stringent metric from recent literature. Strict reports the same English Strict Risk-Aware MF_β scores as the main leaderboard, showing stronger discriminative power by penalizing models that make fatal errors more severely.

contrast, *hard errors* are intentionally designed as plausible near-miss alternatives: they may rely on partially correct evidence or otherwise resemble valid reasoning, but ultimately remain incorrect. We therefore penalize easy errors more heavily than hard errors.

Importantly, the fixed weights were determined from the benchmark's annotation design before model-level analysis, rather than tuned to maximize any particular model ranking. Our goal is not to favor a specific system, but to reflect the benchmark's intended risk structure: options that more clearly violate chart evidence should incur larger penalties.

D.3 Formal Definition

Let A be the set of correct answers (Ground Truth) and B be the set of answers selected by the model. The incorrect options are categorized into two subsets: Easy/Obvious Errors (E) and Hard/-Confusing Errors (H). We define:

$$TP = |A \cap B|, \quad e = |B \cap E|, \quad h = |B \cap H| \tag{5}$$

The **Net True Positive** with fixed severity weights is:

$$TP^{\star}_{net} = \max(0,\ TP - w_e \cdot e - w_h \cdot h) \tag{6}$$

The fixed severity weights are:

$$w_e = 1.0, \qquad w_h = 0.5 \tag{7}$$

Here, “easy” and “hard” refer to distractor discriminability rather than evaluation importance. The larger penalty on easy errors reflects the fact that these options correspond to clearer grounding failures or explicitly incorrect conclusions, whereas hard errors are designed to be more plausible alternatives.

Based on $TP^{\star}_{net}$, we define:

$$P^{\star}_{strict} = \begin{cases} 0, & |B| = 0 \\ \frac{TP^{\star}_{net}}{|B|}, & |B| > 0 \end{cases} \qquad R = \begin{cases} 0, & |A| = 0 \\ \frac{|A \cap B|}{|A|}, & |A| > 0 \end{cases} \tag{8}$$

The final $MF_\beta^{strict\star}$ score is:

$$MF_\beta^{strict\star} = \begin{cases} 0, & \beta^2 P^{\star}_{strict} + R = 0 \\ (1+\beta^2) \cdot \frac{P^{\star}_{strict} \cdot R}{\beta^2 P^{\star}_{strict} + R}, & \text{otherwise} \end{cases} \tag{9}$$

D.4 Interpretation

- When $\beta = 1$, the metric balances precision and recall equally.
- When $\beta > 1$, greater emphasis is placed on avoiding incorrect selections (higher precision).
- When $\beta < 1$, greater emphasis is placed on covering correct options (higher recall).

The *cancellation logic* in $TP^{\star}_{net}$ means that severe errors actively deduct from the model's score rather than merely lowering precision. This mirrors high-stakes decision-making, where a single fatal flaw can render a strategy unusable.

D.5 Why Not Standard F1?

Standard F_1 treats all false positives equally, regardless of whether the selected option is an obvious grounding failure or a plausible near-miss. This is unsuitable for our setting, where distractors are deliberately constructed with different semantic roles and different degrees of risk. The proposed metric preserves the familiar precision–recall structure of F_β while incorporating task-specific error severity through a simple and fixed penalty scheme. As shown in table 4, this makes the metric more discriminative than standard F_1 and more aligned with the benchmark's decision-oriented evaluation objective.

We compare $MF_\beta^{strict\star}$ with Com^2 [37] in section 3.2, where we highlight $MF_\beta^{strict\star}$'s role in model selection for specific scenarios and its feasibility for preference model training.

Analysis of MF_β Curves

As shown in Figures 5 and 6, we plotted the MF_β curves of all models under varying values of β. Overall, the curves exhibit a monotonically increasing trend, indicating that within the current task setting, "selecting all correct options" is significantly more difficult than "avoiding incorrect options." In other words, the models generally perform better at eliminating incorrect options than at fully covering the correct ones. We hypothesize that this phenomenon is related to the characteristics of the hard options: such options are more prone to being selected by the models, yet their penalty weight in the scoring scheme is relatively low, which to some extent "inflates" the overall scores. This effect is particularly pronounced for smaller-parameter models whose performance approaches random selection.

Intersection points. The intersections between the curves carry critical implications: they reveal the trade-offs between different models in terms of "recalling correct options" versus "avoiding incorrect ones," thereby providing guidance for model selection across application scenarios. For example, in recall-oriented tasks, models that perform better before the intersection point should be prioritized, whereas in precision-oriented tasks, models that excel after the intersection point are preferable.

Curve variability. Taking InternVL2-26B as an example, its curve ranks relatively low when β is small, but improves markedly as β increases. This pronounced change highlights the model's substantial variability across different evaluation emphases, reflecting a lack of balance—that is, an insufficiently stable ability to reconcile recall and precision.

D.6 Language-Specific Results for Task 3 and Task 4

Table 5 presents the detailed Task 3 (Anomaly/Pattern Analysis) and Task 4 (Strategy Recommendation) results across Chinese (cn), English (en), and Spanish (es) for the multi-select setting, together with generative scores kept consistent with the main leaderboard.

Key observations from this breakdown are as follows:

- **Cross-lingual variation in multi-select**: Multi-select performance on Task 3 varies across languages for several models, indicating that anomaly and pattern analysis remains sensitive to language-specific reasoning conditions.
- **Task 4 strategic reasoning**: Strategy recommendation remains difficult in the multi-select setting, while the generative scores provide a complementary view that is directly aligned with the main leaderboard.
- **Open-weight competitiveness**: Qwen3-VL-32B and InternVL3-78B remain among the strongest open-weight models on Task 3 and Task 4, consistent with the main experiments.

D.7 Generative Evaluation for Task 3 and Task 4

To complement the multiple-choice setting, we additionally evaluate Task 3 (*Anomaly and Pattern Analysis*) and Task 4 (*Strategy Recommendation*) in a generative setting. In this setting, models are given the chart images and the question only, without access to the candidate options, and are required to produce a free-text analysis. Our goal is to evaluate both whether the generated analysis implies the correct decision and whether the accompanying reasoning is faithful, complete, and grounded in the chart evidence.

D.7.1 Evaluation Pipeline. The generative evaluation consists of four stages:

- **Stage 1: Generative inference.** The model receives chart images and the question, and produces a free-text analysis.
- **Stage 2: Answer extraction.** A judge LLM reads the generated analysis together with the original multiple-choice question and extracts the option(s) implied by the response.
- **Stage 3: Risk-aware option-level scoring.** The extracted option set is evaluated against the original annotations using the same Strict Risk-Aware MF_β protocol as in the multiple-choice setting.
- **Stage 4: Explanation-quality evaluation.** We further assess the generated explanation with an LLM judge and report reference-based similarity metrics as supplementary indicators.

This design keeps the primary generative score directly comparable to the original multiple-choice benchmark, while also providing a richer analysis of explanation quality.

D.7.2 Answer Extraction and Risk-Aware Scoring. Given a free-text response, we use a judge LLM to determine which option(s) the response implicitly endorses. The judge is provided with the original question and the generated analysis, and is asked to output the corresponding option letters. If the response does not clearly support any option, the judge outputs `NONE`.

Listing 1: Answer extraction prompt template

```
Given the following question and a candidate's
    textual analysis,
determine which option(s) the candidate is
    suggesting as the answer.

Question: {question}

Candidate's Analysis:
```

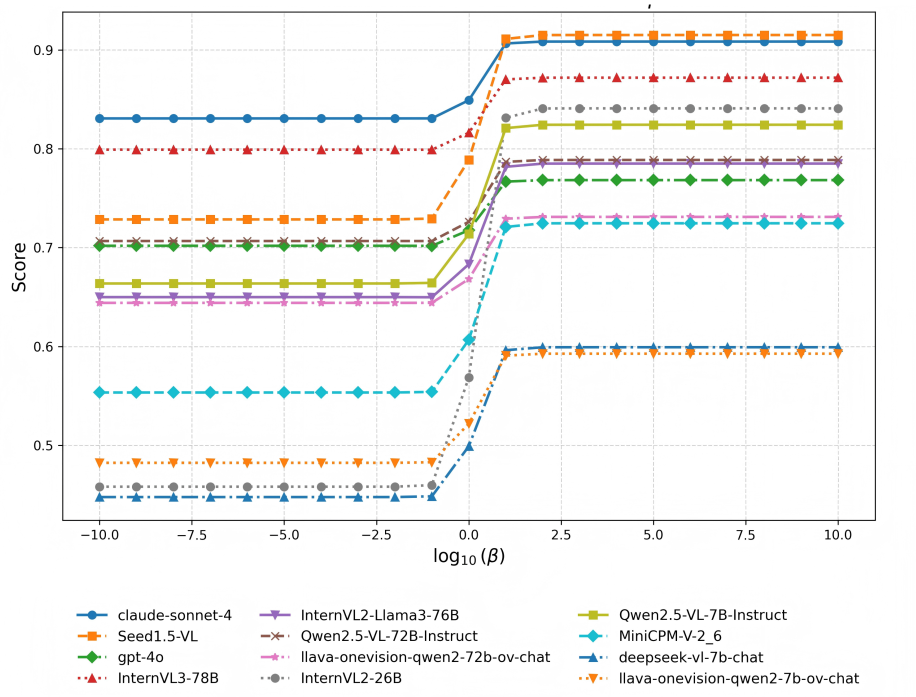

Figure 5: Model Performance on Task 3 under Different β

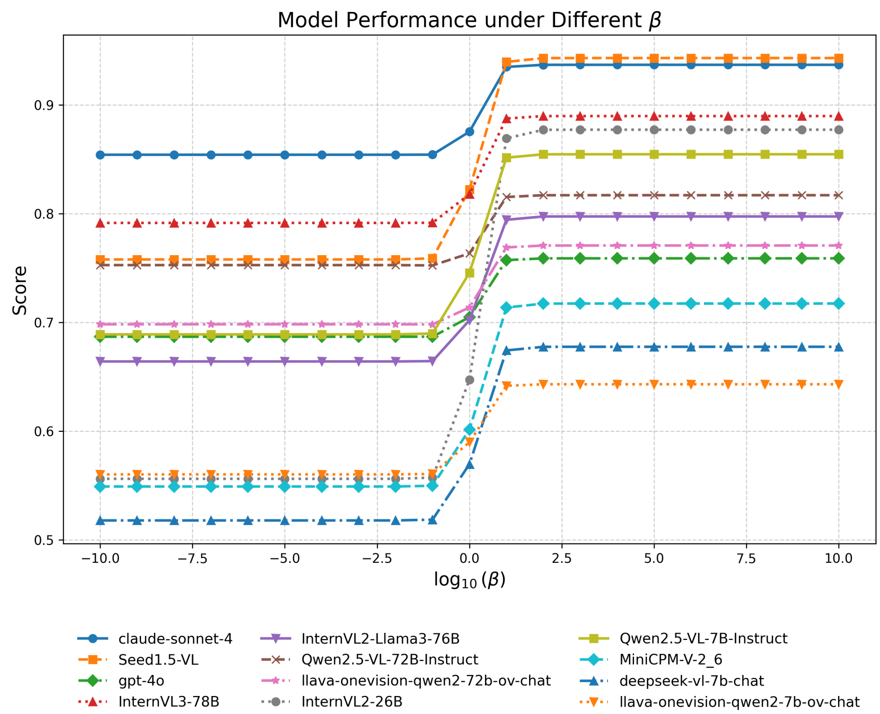

Figure 6: Model Performance on Task 4 under Different β

	Task 3: Anomaly/Pattern Analysis				Task 4: Strategy Recommendation			
	Multi-select			Generative	Multi-select			Generative
Model	cn	en	es		cn	en	es	
Human	**90.60**	**90.60**	**90.60**	**85.80**	**91.60**	**91.60**	**91.60**	**87.50**
Proprietary Models								
Claude-Sonnet-4	75.78	70.00	69.33	**77.6**	77.49	77.49	77.49	73.1
Gemini-2.5-Pro	**79.33**	**75.06**	**75.11**	75.5	75.54	75.54	75.54	**79.7**
Seed1.5-VL	71.78	72.44	67.33	62.8	**78.40**	**78.40**	**78.40**	74.4
GPT-4o	62.64	64.21	59.87	58.2	56.16	56.16	56.16	73.2
Open-weight Models								
InternVL3-78B	68.46	**73.21**	64.66	64.2	**73.00**	**73.00**	**73.00**	61.8
Qwen3-VL-32B	**73.94**	73.11	**72.67**	**76.3**	68.35	68.35	68.35	**77.8**
Qwen3-VL-8B	64.44	62.36	60.44	64.2	70.75	70.75	70.75	73.7
InternVL2-L3-76B	48.88	59.91	59.19	62.9	53.51	58.54	51.14	69.7
Qwen2.5-VL-72B	56.95	56.25	53.13	70.7	63.57	63.57	63.57	76.3
LLaVA-OV-72B	57.59	61.33	53.33	54.8	52.85	52.85	52.85	58.4
InternVL2-26B	58.65	54.46	50.11	47.0	51.22	51.22	51.22	47.0
Qwen2.5-VL-7B	54.91	54.44	52.22	61.7	62.21	62.21	62.21	65.6
MiniCPM-V-2_6	55.36	49.88	44.22	54.1	41.23	41.23	41.23	59.6
DeepSeek-VL-7B	47.11	49.32	40.44	50.5	29.32	29.32	29.32	53.0
LLaVA-OV-7B	34.00	48.55	46.00	49.9	38.77	38.77	38.77	58.5

Table 5: Task 3 (Anomaly/Pattern Analysis) and Task 4 (Strategy Recommendation) results breakdown by language (cn, en, es). Multi-select columns report Strict Risk-Aware MF_β ($\beta = 1$, $w_e = 1.0$, $w_h = 0.5$), and Generative columns report the same free-form generation scores as the main leaderboard. Best scores in each column are in bold; "-" indicates unavailable results.

```
{generated_text}

Extract the option letter(s) (e.g., A, B, C, D
    , E, F) that the
analysis implies as the correct answer.
If the analysis does not clearly indicate any
    option, respond with:
NONE
```

After answer extraction, we evaluate the predicted option set against the original task annotations, including the correct labels as well as the *easy-error* and *hard-error* option groups. We then compute the same Strict Risk-Aware MF_β score as in the multiple-choice setting. In this way, the primary generative score remains aligned with the benchmark's original risk-aware evaluation principle, rather than relying solely on free-form semantic similarity.

D.7.3 Explanation-Quality Evaluation. In addition to answer correctness, we evaluate the quality of the generated explanation with an LLM judge along four dimensions:

- **Evidence Relevance (0–1):** whether the cited chart evidence, values, and trends are directly relevant to the question and final conclusion;
- **Reasoning Completeness (0–1):** whether the reasoning chain covers the key intermediate steps without major omissions;
- **Hallucination Risk (0–1):** whether the explanation avoids introducing unsupported claims that are not grounded in the chart evidence; higher is better;
- **Consistency (0–1):** whether the intermediate statements, reasoning process, and final conclusion are mutually consistent.

The judge is prompted to output structured scores in JSON format:

Listing 2: Explanation-quality judge prompt template

```
You are an expert evaluator for multi-chart
    reasoning explanations.

Evaluate the candidate's explanation based on
    the following rubric.
Output strictly as JSON.

Question: {question}
Gold Correct Answer: {gold_answer}
Gold Explanation: {gold_explanation}

```

```
Candidate's Analysis:
{candidate_analysis}

Rubric:
- evidence_relevance: Whether cited evidence
    is directly relevant
  to the question and final answer.
- reasoning_completeness: Whether the
    reasoning chain covers key
  steps without major omissions.
- hallucination_risk: Whether the explanation
    introduces unsupported
  facts not grounded in chart evidence.
- consistency: Whether reasoning, intermediate
     statements, and final
  answer are mutually consistent.

Output format:
{
  "evidence_relevance": float (0-1),
  "reasoning_completeness": float (0-1),
  "hallucination_risk": float (0-1),
  "consistency": float (0-1),
  "overall": float (0-1),
  "comments": str
}
```

This stage is intended to capture explanation quality beyond discrete option correctness.

D.7.4 Reference-Based Similarity. We further report reference-based similarity metrics between the generated explanation and the gold explanation, including **BLEU-4**, **BERTScore**, and **Rouge-L**. These metrics serve as supplementary indicators of lexical and semantic overlap, but are not used as the primary leaderboard metric, since a valid explanation may differ substantially in wording from the reference while still being correct and well grounded.

D.7.5 Reported Metrics. In the main leaderboard, we report **Strict_F1** as the primary score for generative evaluation, where the predicted answer set is obtained through answer extraction and then evaluated under the same risk-aware option-level protocol as in the multiple-choice setting. We additionally report the four LLM-judge dimensions and the reference-based similarity metrics in supplementary analyses to provide a more complete picture of generative reasoning quality.

E Discussion

E.1 Impact of Irrelevant Charts

In these experiments, because the number of charts varies and high-quality charts with many visual tokens can cause the total input length to exceed the model's context window, the model will begin discarding tokens from the left (i.e., the earliest inputs) until the remaining token count less than session_len. Therefore, we deliberately placed any image tokens for charts unrelated to the QA pair on the far left; even if they are dropped, the model's ability to answer correctly remains unaffected.

Method for handling excessively long image tokens. In these experiments, because the number of charts varies and high-quality charts with many visual tokens can cause the total input length to exceed the model's context window, the model will begin discarding tokens from the left (i.e., the earliest inputs) until the remaining token count less than session_len. Therefore, we deliberately placed any image tokens for charts unrelated to the QA pair on the far left; even if they are dropped, the model's ability to answer correctly remains unaffected.

E.2 Visual Dependency and Language Priors

To rigorously verify whether the models rely on visual perception or exploit textual shortcuts, we conducted a blind ablation study (Table 8) by removing the chart inputs. The results highlight three key findings:

Necessity of Visual Information. In the *Text Only* setting of Task 1 (Trend Inference), models achieve scores around 47%–50%, effectively performing random guessing on the binary task. However, when explicitly instructed to answer "unknown" if charts are missing (*Text+Hint*), accuracy drops precipitously (e.g., Seed1.5-VL drops to 3.56%). Since the ground truth labels are binary (Yes/No), this near-zero score confirms that models successfully ceased random guessing and followed instructions to reject the query. This sharp contrast validates that the high performance in the *Standard* setting is driven by genuine visual perception rather than textual speculation. Similarly, the collapse of performance in Task 2 (Data Integration) without images (e.g., InternVL3-78B: 5.60%) further proves that precise numerical reasoning in our benchmark is strictly vision-dependent.

LLM Priors vs. Visual Grounding. In Task 4 (Strategy Recommendation), models maintain relatively high baselines even in the *Text Only* setting (e.g., Seed1.5-VL: 76.04%). This suggests that advanced LLMs can leverage their internal knowledge bases to propose plausible business strategies based solely on the textual context of the question. However, introducing visual information (*Standard*) consistently yields performance gains (increasing to 78.46%). This demonstrates that while language priors provide a strong foundation for decision-making, visual alignment is essential for tailoring strategies to the specific data trends presented in the charts, thereby achieving the optimal solution.

E.3 Performance Across Different Languages

Robustness in Mixed-Language Scenarios. The results on mixed-language charts (Table 13) further support the "language-agnostic" nature of advanced visual perception. Top-tier models such as Seed1.5-VL and InternVL3-78B maintain performance close to their monolingual baselines even when English, Chinese, and Spanish charts are interleaved within a single query. For example, Seed1.5-VL achieves 70.38% on Trend Inference with mixed charts, close to its English-only baseline of 72.44%. This suggests that SOTA models handle multi-chart QA not by relying on one unified linguistic context, but by performing semantic extraction on each chart regardless of source language, showing strong generalization in heterogeneous real-world settings.

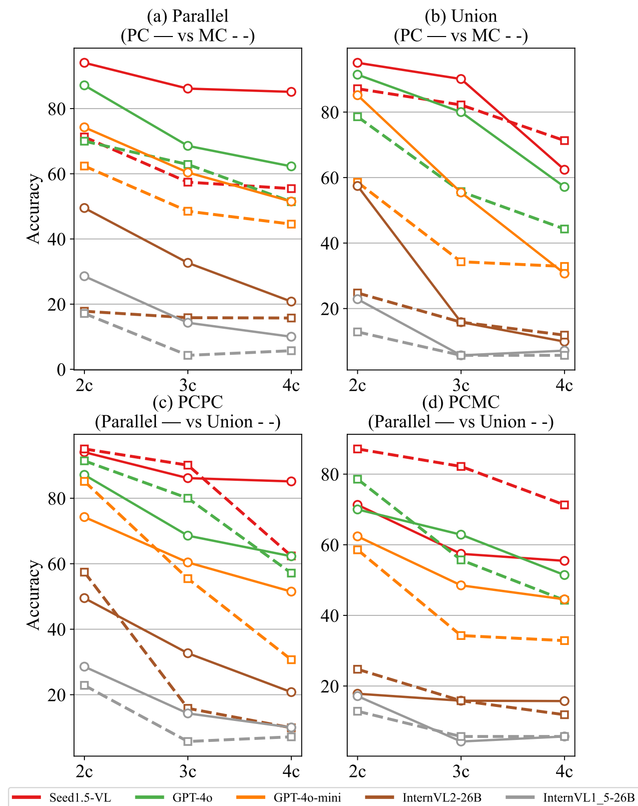

Figure 7: Compare the performance of MLLMs across four retrieval tasks. The x-axis represents the number of charts involved in the question, for example, 2c denotes two charts.

E.4 Exploring MLLMs' Retrieval Capabilities

E.5 Error Analysis

The detailed response statistics of the four proprietary models on Task 3 and Task 4 are presented in Tables 14 & 15.

F Extended Benchmark.

F.1 Task Description

parallel type.

- EXAMPLE
  question: one sub-question for chart 1; one for chart 2; one for chart 3; and one for chart 4
  answer: answer1. answer2. answer3. answer4.

Model	Trend Inference		Data Integration		Anomaly/Pattern Attr		Strategy Rec	
	involved	all	involved	all	involved	all	involved	all
Claude-Sonnet-4	70.00	69.11	60.99	61.38	73.99	73.52	78.21	78.31
Seed1.5-VL	72.44	72.83	67.66	66.74	68.85	68.26	73.84	71.50
GPT-4o	64.21	66.14	64.83	63.10	49.05	47.18	48.96	49.08
InternVL3-78B	73.21	60.93	67.50	64.68	66.28	65.68	65.68	63.05
InternVL2-Llama3-76B	59.91	52.46	51.59	44.36	46.38	45.22	47.24	46.86
Qwen2.5-VL-72B-Instruct	56.25	58.93	25.40	22.94	48.23	50.33	52.18	48.93
LLaVA-OV-72B	61.33	59.11	43.02	38.07	40.76	40.29	43.21	42.73
InternVL2-26B	54.46	46.33	31.03	22.90	47.60	42.82	48.96	48.85
Qwen2.5-VL-7B-Instruct	44.00	53.56	21.04	23.98	51.93	45.11	53.30	49.28
MiniCPM-V-2_6	49.88	52.89	23.29	27.48	33.81	35.85	30.93	32.78
DeepSeek-VL-7B	49.32	24.22	7.95	3.48	21.23	16.19	24.09	23.87
LLaVA-OV-7B	48.55	45.19	21.84	13.55	19.15	19.81	22.88	19.74

Table 6: Compare the question-answering performance when inputting all charts versus only the relevant charts.

Model	Trend Inference		Data Integration		Anomaly/Pattern Attr		Strategy Rec	
	involved	all	involved	all	involved	all	involved	all
Claude-Sonnet-4	70.00	69.11	60.99	61.38	65.04	62.56	69.67	69.15
Seed1.5-VL	72.44	72.83	67.66	66.74	58.72	58.63	66.30	63.81
GPT-4o	64.21	66.14	64.83	63.10	39.29	35.55	39.13	38.54
InternVL3-78B	73.21	60.93	67.50	64.68	57.37	54.25	57.20	54.20
InternVL2-Llama3-76B	59.91	52.46	51.59	44.36	34.24	33.99	36.95	36.18
Qwen2.5-VL-72B-Instruct	56.25	58.93	25.40	22.94	32.93	33.19	39.44	37.26
LLaVA-OV-72B	61.33	59.11	43.02	38.07	27.26	27.48	31.67	30.91
InternVL2-26B	54.46	46.33	31.03	22.90	36.32	30.81	36.53	37.27
Qwen2.5-VL-7B-Instruct	44.00	53.56	21.04	23.98	37.33	32.81	43.19	39.52
MiniCPM-V-2_6	49.88	52.89	23.29	27.48	21.27	24.63	21.90	22.86
DeepSeek-VL-7B	49.32	24.22	7.95	3.48	11.35	7.59	17.19	16.30
LLaVA-OV-7B	48.55	45.19	21.84	13.55	9.20	8.97	12.03	10.13

Table 7: Compare the question-answering performance when inputting all charts versus only the relevant charts under the com2 metric.

Model	Trend Inference			Data Integration		Anomaly/Pattern Attr		Strategy Rec	
	Text+Hint	Text Only	Standard	Text Only	Standard	Text Only	Standard	Text Only	Standard
Seed1.5-VL	3.56	47.33	71.78	6.31	72.81	63.76	75.35	76.04	78.46
InternVL3-78B	14.51	47.44	68.46	5.60	70.72	51.46	66.38	53.94	68.24
MiniCPM-V-2_6	21.56	46.00	55.36	4.66	26.15	26.90	31.44	27.33	35.02

Table 8: Ablation study on visual modality dependency. Standard: The model receives both chart images and text prompts. Text Only: The model receives only text prompts without chart images. Text+Hint: In Task 1, the model receives only text but is explicitly instructed to output ”unknown” if visual information is missing. Note that in the ’Text+Hint’ setting, an output of ”unknown” is counted as incorrect (0 score) against the ground truth (Yes/No), strictly penalizing non-visual guessing.

Model	Trend Inference			Data Integration			Anomaly/Pattern Attr			Strategy Rec		
	en	zh	es	en	zh	es	en	zh	es	en	zh	es
Seed1.5-VL	72.44	71.56	67.56	67.66	68.76	67.04	68.85	72.78	68.80	73.84	77.52	71.41
InternVL3-78B	73.21	70.98	66.00	67.50	70.69	65.91	66.28	68.10	62.04	65.68	67.32	62.48
InternVL2-26B	54.46	46.09	45.07	31.03	28.57	21.72	47.60	38.85	37.54	48.96	42.59	39.82
MiniCPM-V-2_6	49.88	51.79	48.78	23.29	29.50	23.95	33.81	36.99	27.53	30.93	35.12	28.61
Qwen2.5-VL-7B	54.44	55.23	48.67	21.04	23.81	23.29	51.93	48.29	50.82	53.30	47.11	60.56
LLaVA-OV-7B	48.55	33.48	46.33	21.84	18.26	13.14	19.15	22.14	30.73	22.88	32.11	27.77

Table 9: English Charts - Different Language QAs

Model	Trend Inference			Data Integration			Anomaly/Pattern Attr			Strategy Rec		
	en	zh	es	en	zh	es	en	zh	es	en	zh	es
Seed1.5-VL	72.44	68.22	69.78	67.66	65.69	65.70	68.85	71.86	70.75	73.84	74.36	71.66
InternVL3-78B	73.21	68.97	69.87	67.50	67.26	58.03	66.28	66.24	64.55	65.68	65.62	66.36
InternVL2-26B	54.46	53.56	48.44	31.03	18.20	19.46	47.60	46.80	42.95	48.96	53.25	47.83
MiniCPM-V-2_6	49.88	50.00	49.56	23.29	24.53	25.29	33.81	36.41	36.05	30.93	31.79	30.78
Qwen2.5-VL-7B	54.44	52.35	52.67	21.04	14.41	20.95	51.93	48.70	53.74	53.30	49.16	53.56
LLaVA-OV-7B	48.55	41.67	46.85	21.84	5.11	14.25	19.15	18.75	19.92	22.88	17.95	20.32

Table 10: Different Language Charts - English QAs

Model	Trend Inference			Data Integration			Anomaly/Pattern Attr			Strategy Rec		
	en	zh	es	en	zh	es	en	zh	es	en	zh	es
Seed1.5-VL	72.44	71.56	67.56	67.66	68.76	67.04	58.72	63.57	59.39	66.30	69.63	62.52
InternVL3-78B	73.21	70.98	66.00	67.50	70.69	65.91	57.15	55.21	51.69	57.20	58.60	52.78
InternVL2-26B	54.46	46.09	45.07	31.03	28.57	21.72	36.32	26.54	26.76	36.53	31.22	29.44
MiniCPM-V-2_6	49.88	51.79	48.78	23.29	29.50	23.95	21.27	25.93	17.44	21.90	24.93	18.01
Qwen2.5-VL-7B	54.44	55.23	48.67	21.04	23.81	23.29	37.33	35.05	39.07	43.19	36.07	48.67
LLaVA-OV-7B	48.55	33.48	46.33	21.84	18.26	13.14	9.20	11.19	18.52	12.03	20.04	17.04

Table 11: English Charts - Different Language QAs Under the Com2 Metric

Model	Trend Inference			Data Integration			Anomaly/Pattern Attr			Strategy Rec		
	en	zh	es	en	zh	es	en	zh	es	en	zh	es
Seed1.5-VL	72.44	68.22	69.78	67.66	65.69	65.70	58.72	63.06	61.43	66.30	66.00	63.52
InternVL3-78B	73.21	68.97	69.87	67.50	67.26	58.03	57.15	56.34	53.75	57.20	55.47	57.23
InternVL2-26B	54.46	53.56	48.44	31.03	18.20	19.46	36.32	35.51	30.54	36.53	41.07	35.94
MiniCPM-V-2_6	49.88	50.00	49.56	23.29	24.53	25.29	21.27	23.19	23.04	21.90	21.86	21.41
Qwen2.5-VL-7B	54.44	52.35	52.67	21.04	14.41	20.95	37.33	35.26	41.41	43.19	39.20	43.52
LLaVA-OV-7B	48.55	41.67	46.85	21.84	5.11	14.25	9.20	8.43	8.75	12.03	8.48	11.15

Table 12: Different Language Charts - English QAs Under the Com2 Metric

- PCPC

A question contains multiple parallel sub-questions, each retrieving one piece of information from a single chart.

Model	Trend Inference	Data Integration	Anomaly/Pattern Attr	Strategy Rec
Seed1.5-VL	70.38	65.11	69.54	74.54
InternVL3-78B	69.93	64.96	68.23	66.18
Qwen2.5-VL-7B-Instruct	54.12	19.38	49.94	51.17

Table 13: The test results of English Q&A on mixed-language charts.

Option	claude	seed	gpt	gemini
A	24.5 / 6.3	36.8 / 0.8	64.2 / 1.3	26.9 / 1.7
B	7.9 / 9.2	10.2 / 8.1	3.4 / 48.0	2.3 / 19.0
C	9.6 / 10.3	6.9 / 12.6	4.8 / 53.4	2.7 / 24.4
D	17.3 / 6.6	36.1 / 0.8	76.0 / 0.8	36.1 / 1.2
E	9.4 / 11.6	9.9 / 10.5	4.7 / 51.6	1.6 / 23.3
F	8.9 / 5.6	29.7 / 1.6	63.4 / 1.6	34.7 / 1.6
G	12.4 / 4.3	33.8 / 1.6	57.9 / 1.0	26.9 / 0.7

Table 14: Omission and multiple-selection rates per option for each model for task 3 (%). In this context, "claude" refers to Claude-Sonnet-4, "seed" refers to Seed1.5-VL, "gpt" refers to GPT-4o, and "gemini" refers to Gemini-2.5-Pro.

Option	claude	seed	gpt	gemini
A	11.6 / 4.4	40.7 / 1.2	62.8 / 0.4	31.6 / 0.5
B	4.3 / 9.1	2.9 / 7.9	1.9 / 49.2	3.0 / 21.1
C	2.5 / 6.8	2.5 / 10.4	1.0 / 47.4	1.2 / 30.5
D	8.9 / 2.7	32.8 / 0.0	62.0 / 1.2	25.2 / 0.9
E	4.2 / 11.6	7.8 / 12.8	2.1 / 58.1	0.0 / 32.7
F	9.1 / 4.4	25.6 / 0.7	58.0 / 2.9	26.3 / 3.8
G	7.3 / 4.0	35.8 / 0.0	58.9 / 2.3	34.2 / 0.0

Table 15: Omission and multiple-selection rates per option for each model for task 4 (%). In this context, "claude" refers to Claude-Sonnet-4, "seed" refers to Seed1.5-VL, "gpt" refers to GPT-4o, and "gemini" refers to Gemini-2.5-Pro.

2c, 3c, and 4c refer to retrieving one piece of information from charts 2, 3, and 4, respectively, and providing separate answers.

- PCMC
  A question contains multiple parallel sub-questions, each retrieving multiple pieces of information from a single chart.
  2c, 3c, and 4c refer to retrieving at least one piece of information from charts 2, 3, and 4, respectively, and providing separate answers.

union type.

- EXAMPLE
  question: question about chart_1&2&3&4
  answer: answer.
- PCPC
  The question involves multiple charts, retrieving one piece of information from each chart, combining the information to perform a simple calculation, and outputting the final result.
- PCMC
  The question involves multiple charts, retrieving at least one piece of information from each chart, combining the information to perform a simple calculation, and outputting the final result.

F.2 Pipeline

1. Manually select articles from Pew that contain at least four charts, and scrape both the articles and charts.
2. Extract chart information and perform manual sampling-based screening to ensure quality.
3. Automatically generate topic summaries based on the content of the article.
4. Generate question-and-answer pairs by utilizing the extracted chart information and setting different prompts based on the topics.
5. Use scripts to check whether the number of charts involved in all question-and-answer pairs and the amount of content related to each chart meet the expected criteria. If they do not meet the criteria, regenerate the pairs and manually

Category	Number of Articles
Economy & Work	11
Politics & Policy	11
Internet & Technology	11
Family & Relationships	11
Age & Generations	11
Immigration & Migration	11
Science	11
News Habits & Media	12
Other Topics	12

Table 16: Category-wise Number of Articles

adjust them until all question-and-answer pairs satisfy the conditions.

6. Use scripts to check the length of all answers, manually screen answers that exceed the threshold, remove redundant parts, and retain concise answers.
7. Utilize the rationale of generated question-and-answer pairs to generate code-based computation results, manually compare all inconsistent answers, correct the answers and rationales in the labels, and ensure the accuracy of the computation results.

F.3 Data Statistics

We collected 101 articles from Pew Research Center, each containing at least four charts centered on the same topic, spanning nine topics in total (Refer to Table 16), yielding 1,212 QA pairs. These QA pairs span four distinct question types, comprehensively probing models' capabilities in cross-chart information retrieval.

G QAs Prompts

For Task 3 and Task 4, we use neutral multiple-choice prompt templates that do not contain one-shot answer demonstrations or fixed label patterns, in order to reduce prompt-induced option bias.

Task 1: Trend Inference

```
output_prompt = '''
    Output according to the following json format:
    {
        "rationale": "Reasoning process",
        "answer": "Final answer"
    }
    Please strictly output according to the json format.
'''
prompt = image_tokens + question + output_prompt
```

Task 2: Data Integration

```
output_prompt = '''
    Please answer according to the following steps:
        1. Extract relevant information...
        2. Explain the association logic...
        3. Calculation process...
        4. Conclusion summary...
        5. Output requirements...
'''
prompt = image_tokens + question + output_prompt
```

Task 3: Anomaly/Pattern Attribution

```
text1 = "The options for the question are as follows: "
hints = "(Multiple choice question) Please analyze..."
output_prompt = '''
    Answer the question according to the following JSON format:
    {
        "rationale": "Reasoning process grounded in the chart evidence",
        "answer": ["option_letter_1", "option_letter_2"]
    }
    The "answer" field should contain only the option letters you judge to be correct.
    Do not imitate any fixed option pattern; determine the answer solely from the chart
        content.
'''
prompt = image_tokens + hints + question + text1 + answer_choices + output_prompt
```

Task 4: Strategy Recommendation

```
text1 = "The options for the question are as follows: "
hints = "(Multiple choice question) Please analyze the content of the chart and select the
    correct options based on the chart information. Note: Correct options should be supported
    by chart data; otherwise, they will be regarded as incorrect."
output_prompt = '''\nAnswer the question according to the following JSON format:
    {
        "rationale": "Reasoning process grounded in the chart evidence",
        "answer": ["option_letter_1", "option_letter_2"]
    }
    Select the correct option letter(s) according to the actual chart content.
    Do not follow any fixed answer pattern; use only chart-supported evidence.
'''
prompt = image_tokens + hints + question + text1 + answer_choices + output_prompt
```

The prompt for generating python calculation code based on the reasoning process in task 2 is as follows:

```
gen_code_prompt = '''
    The above is the reasoning process. Since large models are not good at calculations,
        ignore the calculated results in the reasoning process above, and generate executable
        Python code for the reasoning process.
    Please note:
    1. Strictly generate Python code based on the "rationale" process above.
    2. Ensure the code correctly reflects the reasoning of the "rationale" by using variables
        to replace the intermediate calculation results in the rationale, as the code
        execution is more accurate and avoids cumulative errors caused by using intermediate
        calculation results.
    3. The final output should strictly follow the required format, only printing the last
        output from the final `print` statement, without any descriptive print statements.
        Below are some examples of the final print outputs for reference:
        a. Output format: "The answer should be presented as an integer."
            Incorrect final output 1: print(f"The final answer is: {answer}.")
            Incorrect final output 2: print(f"**{answer}**")
            Correct final output: print(f"{answer}")
        b. Output in percentage, rounded to 4 significant digits.
            Incorrect final output 1: print(f"In mobile search, the proportion of images is
                **{answer}%**.")
            Correct final output: print(f"{answer}%")
        c. Output in dollars, rounded to 3 decimal places. Answer: xx dollars.
            Incorrect final output 1: print(f"In Q1 2023, the in-app purchase revenue per
                download for mobile games on the App Store in Japan is {answer} USD.")
            Correct final output: print(f"{answer} USD")
        d. Output "Yes" or "No".
            Incorrect final output 1: print("True")
            Correct final output: print("Yes")
        e. Answer: xx times.
            Incorrect final output 1: print(f"{answer}")
            Correct final output: print(f"{answer} times")
    4. Be sure to distinguish between "significant digits" and "decimal places." "Significant
        digits" refer to all digits from the first non-zero digit to the last digit. "Decimal
        places" refer to the number of digits after the decimal point.
    5. If floating-point calculations are involved, please use the `decimal` library to avoid
        precision loss from floating-point operations.
'''
```